

THE PROBLEM OF THE TIDES

WITH REGARD TO OSCILLATIONS OF THE SECOND KIND.

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It is known to those who are conversant with LA PLACE'S theory of the tides, that, in the solution of the problem with regard to oscillations of the second kind, he obtained the remarkable result, that if the whole surface of the earth were covered by a sea of uniform depth, the diurnal tides, which are produced by these oscillations, would vanish. Although LA PLACE seems to have regarded this as one of his most happy discoveries, and notwithstanding Professor AIRY has since arrived at the same result by a somewhat different analysis (*Encyclopædia Metropolitana, Tides and Waves*, p. 102), yet we are persuaded that the solutions of both these distinguished mathematicians are defective, and that the results which they have obtained are erroneous, or at least not applicable to the problem when properly restricted, or even as restricted by LA PLACE himself. We propose, therefore, to point out, in the first place, what we conceive to be defects in these solutions, and to give some reasons for considering the results erroneous; and, secondly, to give what we consider a more satisfactory solution of the problem. In doing this, we must, for the sake of brevity, avail ourselves as much as possible of what has been given on the subject, and hence frequent reference will be made to BOWDITCH'S Commentary on the *Mécanique Céleste*.

LA PLACE has correctly obtained the three equations (2175—2176'), which express the conditions of the equilibrium of the forces, and which must be satisfied in the solution of the problem; and these are substantially the same as those obtained by Professor AIRY (*Tides and Waves*, p. 87), differing only in form and notation. As the solution of the problem in its most general form is impracticable, both LA PLACE and Professor AIRY limit themselves to the case in which the depth of the sea is a function of the latitude only; and after substituting for y, u, v , &c., their assumed values in (2178—2178'''), and eliminating b and c , LA PLACE obtains the equation (2183). With regard to this equation he says (2191) that "*in the general case where n does not vanish, and the depth of the sea is variable, the integration of the equation (2183) surpasses the power of analysis; but to determine the oscillations of the ocean, it is not necessary to integrate it generally, since it is sufficient merely to satisfy it.*" Accordingly, he assumes that the equation can be satisfied by the function $a = Q\mu\sqrt{1-\mu}$ (2144), Q being independent of μ , and finds that it will satisfy it when $Q = 0$, which makes the tide vanish. But in questions involving the conditions of the equilibrium of forces, there may sometimes be more than one result which satisfies the

conditions, as might be shown in numerous cases, only one of which gives a stable equilibrium of the forces, or the problem may be indeterminate, as will be shown in this case, in which an infinite number of results may satisfy the conditions, only one of which may be applicable to the problem where friction or any disturbing influences are concerned. We think, therefore, that the principle above, laid down by LA PLACE, is a little too broad, and that it is *not sufficient* merely to satisfy the equation; and consequently that his solution is defective, inasmuch as there is nothing in his analysis by which to determine whether the function above, although it satisfies the equation, may not give an unstable equilibrium only of the forces, or whether there may not be other functions which will also satisfy the equation, and whether the consequent motions of the sea, as deduced from all of them, may not depend upon a primitive impulse of the sea, and not upon the action of the disturbing body. For LA PLACE'S analysis only shows that the sea might be induced with certain oscillatory motions produced by some primitive impulse, which would satisfy the equation of continuity (2175), and which would also give such values to the first two terms of (2176, 2176') as would exactly counteract the tangential forces of the disturbing body expressed by the last terms; for in this case the tide must vanish, and consequently the third terms. But if the motions of the sea which satisfy these equations upon the hypothesis that $a = 0$, depend upon primitive impulse, the results deduced from such an hypothesis, although the equations are satisfied, must be rejected as inapplicable to the problem; for, as LA PLACE himself says (2191''), such motions must quickly disappear by the resistances of various kinds which the waters of the sea suffer in their motions. And that the results which LA PLACE has obtained do depend upon motions of this kind may be shown without going into an elaborate investigation of the subject, by applying them to the case in which n or the rotatory motion of the earth approximates to 0; for a result can frequently be shown to be erroneous more conveniently and satisfactorily by showing that it fails in extreme cases, than in any other way. Now if the motions of the sea which satisfy the equations for LA PLACE'S result of $a = 0$, do not depend upon a primitive impulse independent of the action of the disturbing body, the same must be true for all values of n , however small; for all the equations were obtained upon the hypothesis of small motions. Hence, as n approximates to 0, that is, as the rotatory motion of the earth approximates to a state of rest, the motions of the sea should also approximate to that state.

But it may be seen by examining the equations of LA PLACE (2260, 2261), which express the motions of the sea, that as n approximates to 0, u and v , and consequently $\frac{du}{dt}$ and $\frac{dv}{dt}$, which express the velocities of these motions, approximate towards infinity. There must then be other results which satisfy the equations than those which satisfy the equations when $a = 0$, which make the motions of the sea vanish as n vanishes; and it is only when the problem is restricted to the condition that $a = 0$ that larger motions are required to satisfy the equations as n decreases.

We shall now endeavor to give a direct solution of the problem which will verify what has been stated, and which is not only applicable to the particular case in which $i = n$, but holds good for any value of i , and for any law of the depth, expressed by a function which admits of being developed into a series containing the even powers of the sines and cosines of the latitude. We shall commence the solution with LA PLACE's fundamental equation (2183), changing a little its notation. If in this equation we use $\cos \theta$ for μ , and restore the value of z in (2182), remembering that $s = 1$ in oscillations of the second kind, it may be reduced to the following form:

$$\frac{i \sin \theta \cdot a}{g} = \frac{d}{d\theta} \left\{ \gamma \frac{2n \cos \theta a' + i \sin \theta \frac{da'}{d\theta}}{4n^2 \cos^2 \theta - i^2} \right\} - \gamma \cdot \frac{2n \sin \theta \cos \theta \frac{da'}{d\theta}}{\sin \theta (4n^2 \cos^2 \theta - i^2)}. \quad (\text{A.})$$

Differentiating, and multiplying by $\sin \theta (4n^2 \cos^2 \theta - i^2)^2$, it becomes

$$\begin{aligned} \frac{i \sin^2 \theta}{g} (4n^2 \cos^2 \theta - i^2)^2 a &= \frac{d}{d\theta} \gamma (4n^2 \cos^2 \theta - i^2) i \sin^2 \theta \\ &+ \frac{d}{d\theta} \left\{ \gamma (8n^2 \sin^2 \theta + 4n^2 \cos^2 \theta - i^2) i \sin \theta \cos \theta + \frac{d}{d\theta} \gamma (4n^2 \cos^2 \theta - i^2) i \sin^2 \theta \right\} \\ &+ a' \cdot \gamma (i^3 + 2n i^2 \sin^2 \theta - 4n^2 i \cos^2 \theta + 8n^3 \sin^2 \theta \cos^2 \theta) \\ &+ a' \cdot \frac{d}{d\theta} \gamma (4n^2 \cos^2 \theta - i^2) 2n \sin \theta \cos \theta \\ &- a' (4n^2 \cos^2 \theta - i^2) \sin^2 \theta. \end{aligned} \quad (\text{B.})$$

Again, from (2178), (2178'''), and (2223), using only the part of V' depending on the action of the disturbing body, and putting $K = \frac{\alpha k}{g}$ we obtain

$$a = K \sin \theta \cos \theta + a'. \quad (\text{C.})$$

We shall omit the small part of V' depending upon the at-

traction of the stratum of water by which the sea is elevated above, or depressed below, the general level, as Professor AIRY has done, and also LA PLACE (2280''') in his solution for oscillations of the third kind.

Combining the last equation with the preceding, putting $i = en$, and $\gamma = l(1 + q \sin^2 \theta + r \sin^4 \theta + \&c.)$, we get

$$\begin{aligned} \frac{n^2}{g l} (K \sin \theta \cos \theta + a') (4 - e - 4 \sin^2 \theta)^2 \sin^2 \theta &= \frac{d}{d\theta} \left\{ (4e - e^3) \sin^2 \theta + (4e - 4eq + e^3 q) \sin^4 \theta + \&c. \right\} \\ &+ \frac{d}{d\theta} e \sin \theta \cos \theta \left\{ \begin{aligned} &4 - e^2 + (4 + 4q + 8eq - e^2 q - 2e^3 q) \sin^2 \theta \\ &+ (4q - 8eq + 4r + 16er - e^2 r - 4e^3 r) \sin^4 \theta + \&c. \end{aligned} \right\} \\ &+ a' \left\{ \begin{aligned} &-4e + e^3 + (8 + 4e + 2e^2 - 4eq + 16q - 4e^2 q) \sin^2 \theta \\ &+ (-8 + 8q + 4eq - 2e^2 q + 32r - 4er + 8e^2 r - e^3 r) \sin^4 \theta + \&c. \end{aligned} \right\} \end{aligned} \quad (\text{D.})$$

The value of e in this equation differs very little from unity, being only $\frac{1}{365}$ less for the sun, and $\frac{1}{27}$ nearly for the moon.

In order to satisfy this equation we must put

$$a' = A^{(0)} + A \sin \theta + A^{(2)} \sin^3 \theta + A^{(4)} \sin^5 \theta \dots + A^{(m)} \sin^m \theta. \quad (\text{E.})$$

If we substitute this value of a' in the preceding equation, and develop $\cos \theta$, and then compare the coefficients of like powers of $\sin \theta$, we shall obtain equations from which we can deter-

mine $A^{(0)}$, A , $A^{(2)}$, &c., and hence the value of a in equation (C).

For the sake of simplicity, we shall carry out the solution for the particular case only in which $i = n$, and $q = 0$, as it may be readily seen from this how the solution may be extended to the more general cases.

Putting $e = 1$ and $q = 0$, equation (D) becomes

$$\begin{aligned} \frac{n^2}{g l} (K \sin \theta \cos \theta + a) (9 \sin^2 \theta - 24 \sin^4 \theta + 16 \sin^6 \theta) &= \frac{d}{d\theta} 3 \sin^2 \theta - 4 \sin^4 \theta \\ &+ \frac{d}{d\theta} (3 \sin \theta \cos \theta + 4 \sin^3 \theta \cos \theta) \\ &+ a' (-3 + 14 \sin^2 \theta - 8 \sin^4 \theta). \end{aligned} \quad (\text{F.})$$

The value of a' in equation (E) being substituted in this equation, we get, by comparing coefficients as stated above,

$$\begin{aligned}
A^{(0)} &= 0 \\
A &= A \text{ (indeterminate),} \\
A^{(2)} &= -\frac{1}{2} A + \frac{3}{8} \cdot \frac{n^2}{g l} (A + K) \\
A^{(4)} &= -\frac{1}{8} A + \frac{11}{96} \cdot \frac{n^2}{g l} (A + K) + \frac{3}{64} \cdot \frac{n^4}{g^2 l^2} (A + K) \\
A^{(6)} &= -\frac{1}{16} A + \frac{2623}{10368} \cdot \frac{n^2}{g l} (A + K) - \frac{51}{4608} \cdot \frac{n^4}{g^2 l^2} (A + K) + \frac{3}{1024} \cdot \frac{n^6}{g^3 l^3} (A + K);
\end{aligned} \tag{G.}$$

and generally after $A^{(n)}$, putting m successively equal to the odd numbers 7, 9, &c.,

$$\begin{aligned}
3(m+1)(m-1)A^{(m)} &= (7m^2 - 33m + 24)A^{(m-2)} - 4(m-6)(m-3)A^{(m-4)} \\
&\quad + (9A^{(m-2)} - 24A^{(m-4)} + 16A^{(m-6)}) \frac{n^2}{g l} \\
&\quad + \left(9 \cdot \frac{m-6}{m-3} \cdot \frac{m-8}{m-5} \cdots - \frac{1}{2} - 24 \frac{m-8}{m-5} \cdot \frac{m-10}{m-7} \cdots - \frac{1}{2} + 16 \frac{m-10}{m-7} \cdot \frac{m-12}{m-9} \cdots - \frac{1}{2} \right) \frac{n^2}{g l} K.
\end{aligned} \tag{H.}$$

In comparing the coefficients of $\sin \theta$, we get $3A - 3A = 0$, or $A = A$. Hence any assumed value of A will satisfy this equation, and if we use this same value in the other equations in (G, H) in determining $A^{(2)}$, $A^{(4)}$, &c., we shall obtain a result for a' in (E), which will satisfy (F), and consequently the conditions of the problem. The problem is therefore indeterminate, and may be satisfied with any assumed value of A , even when $K = 0$. But as the motions of the sea depend upon a' , as may be seen from LA PLACE's equations (2178', 2178'', 2180, 2181), if we assume A too large, it will give values to u and v larger than those contemplated in determining the fundamental equations, for they were determined upon the hypothesis of small motions, and hence the result would not be correct.

We have said that the problem is indeterminate; but this is the case only when the motions of the sea are made to depend wholly, or in part, upon a primitive impulse. For if we examine the equations (G, H), it may be seen that the result for any assumed value of A would depend in part only upon K , or the disturbing body, and in part upon A , which must in this case represent the primitive impulse by which the sea received such motions as would satisfy the conditions of the problem. But as these motions, either wholly or in part, would be independent of the disturbing body, and we can con-

ceive of no circumstances which could produce them, and as they must quickly disappear on account of friction, if they were produced, the problem, as it really exists in nature, becomes entirely determinate. For, in this case, we must put $A = 0$, and then the result will depend entirely upon K , and the motions of the sea be reduced to the minimum which can satisfy the conditions of the problem, as they evidently must where friction is concerned.

We shall now determine whether any assumed value of A will cause the tides to vanish, and thus make our result correspond in a particular case with that of LA PLACE. In order to do this, it may be seen from equation (C), that we must satisfy the equation

$$0 = K \sin \theta \cos \theta + a'. \tag{I.}$$

Developing the first term of the last member of this equation, and comparing the coefficients of the same powers of $\sin \theta$, using for a' its value in (E), we obtain

$$\begin{aligned}
A^{(0)} &= 0, \quad A = -K, \quad A^{(2)} = +\frac{1}{2} K, \quad A^{(4)} = +\frac{1}{8} K, \\
A^{(6)} &= +\frac{1}{16} K,
\end{aligned}$$

and generally

$$A^{(m)} = -\left(\frac{m-4}{m-1} \cdot \frac{m-6}{m-3} \cdots - \frac{1}{2} \right) K. \tag{J.}$$

Hence,

$$A^{(m-2)} = -\left(\frac{m-6}{m-3} \cdot \frac{m-8}{m-5} \cdots - \frac{1}{2} \right) K; \text{ and } A^{(m-4)} = -\left(\frac{m-8}{m-5} \cdot \frac{m-10}{m-7} \cdots - \frac{1}{2} \right) K. \tag{K.}$$

If we now put these values of $A^{(0)}$, A , $A^{(2)}$, ..., $A^{(m)}$ respectively equal to those in (G, H), the equations thus obtained will all be satisfied by putting $A = -K$. Hence the result obtained by assuming $A = -K$ will satisfy both the conditions of the problem and the equation (I), which makes the tide vanish, and this is precisely LA PLACE's result. But as any result, as we have seen, except the one obtained by making $A = 0$, makes the motions of the sea depend in part upon a primitive impulse, which must quickly disappear, if it could even receive such impulse, this result is not applicable to the problem, unless it were required merely to determine

what motions the sea must have to produce a merely theoretical equilibrium of the forces, which could not be permanent where there is friction.

If we now take the more general equation (D), and compare the coefficients of $\sin \theta$, as above, we shall get $(4e - e^3)A = 4e - e^3 A$, in which A is again arbitrary and indeterminate, and hence if we assume any value for A , and use it in determining $A^{(2)}$, $A^{(4)}$, &c., we shall obtain a result which will satisfy the conditions of the problem in this case also. But it would be impossible in this case to satisfy the equation (I) with any assumed value of A , and hence, in this more

general case, the tides could not be made to vanish by means of any primitive impulse.

It may be remarked here, that a similar result of indeterminateness was obtained by LA PLACE in his solution for oscillations of the third kind (2288'), which he failed to interpret rightly, and which has been since explained by Professor AIRY (*Tides and Waves*, pp. 111, 112) somewhat as above, except that he furnished no criterion by which to determine what part of the result depends upon the disturbing body, and what upon primitive impulse; and hence left the result entirely indeterminate. According to the interpretation above, LA PLACE in his general equation of the coefficients (2289) should have put $A^{(r+1)} = 0$, which would have simplified the problem very much, and have given the minimum result for the motions of the sea which could satisfy the conditions of the problem,

which is evidently, from what precedes, the only one strictly applicable to the problem. This would have changed his result very much; for instead of a difference of 36 feet (2306') between high and low water at the equator, when the depth of the sea is $5\frac{1}{2}$ miles, it would have given less than one foot, which is more in accordance with observation; for the tides in the Pacific in the open sea are found to be very small.

We think it is therefore evident that *any* assumption that will satisfy LA PLACE's equation (2183) will not furnish a true result, inasmuch as it would make the tides depend upon a primitive impulse; and this is still further evident from the fact, that this equation may be satisfied in this way for oscillations of both the second and third kind, and a result obtained which would give very large tides, even in the case in which the disturbing force vanishes.

Nashville, Tenn., 1856, May 24.

ON THE OBSERVATION OF THE ASTEROIDS.

Professor BRUNNOW, director of the Observatory at Ann Arbor, writes under date of June 3, stating his gratification at the initiative taken by Lieutenant MAURY, in behalf of the Washington Observatory, in carrying out the suggestions for a division of labor in observing the small planets. He expresses the hope that his observatory will in a very few weeks be in sufficient order to permit the commencement of regular observations with the refractor, and his willingness to undertake, in behalf of the Ann Arbor Observatory, the regular observation of the eight following asteroids:—

<i>Flora,</i>	<i>Proserpina,</i>
<i>Clio,</i>	<i>Calliope,</i>
<i>Astrea,</i>	<i>Euphrosyne,</i>
<i>Metis,</i>	<i>Hebe.</i>

Professor BRUNNOW is at present engaged upon tables of

Cambridge, 1856, June 6.

Clio, and hopes before long to undertake the preparation of tables for yet other members of the asteroid group.

In order to facilitate and hasten this division of labor, so far as regards the observation of the asteroids, the Editor undertakes to guarantee that four others shall be also regularly observed in the United States after the beginning of the year 1857, and selects the four following as least likely to interfere with the preferences of others.

<i>Vesta,</i>	<i>Eunomia,</i>
<i>Iris,</i>	<i>Amphitrite.</i>

These will complete for the Western Continent its contingent of one half the known members of the group; and there is little doubt that steps will soon be taken toward the assumption of an equitable proportion of the labor of computing elements, and providing ephemerides.

G.

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